

Measuring AI Ability to Complete Long Software Tasks

50% Task-Completion Time Horizon Doubles Every ~7 Months

GPT-2 (2019): ~2 seconds → o3 (2025): ~1 hour 50 minutes | $R^2 = 0.97$ | Doubling time $\approx$ 207 days

Thomas Kwa, Ben West, Joel Becker et al. — Model Evaluation & Threat Research (METR)

arxiv.org/abs/2503.14499 · March 2025

A New Metric: 50% Task-Completion Time Horizon

How long a task (by skilled human) can an AI complete with 50% success?

1

Assemble 170-Task Suite

Diverse software & research tasks spanning 1 second to 8 hours of skilled human work time

2

Measure Human Time + AI Success Rate

800+ human baselines (2,529 hrs total); AI agents attempt each task; binary success recorded

3

Fit Logistic Model → 50% Horizon

Logistic regression over task duration yields the duration at which the AI achieves 50% success

Key insight: Anchoring to human task duration makes the metric interpretable across vastly different model generations (GPT-2 to o3).

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: “Which file is a shell script?” Choices: “run.sh”, “run.txt”, “run.py”, “run.md”
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

Three task suites combined; 800+ human baselines totaling 2,529 hours

HCAST

97 tasks · 1 min – 30 hrs · Diverse software tasks

RE-Bench

7 tasks · 8 hrs · Difficult ML research engineering

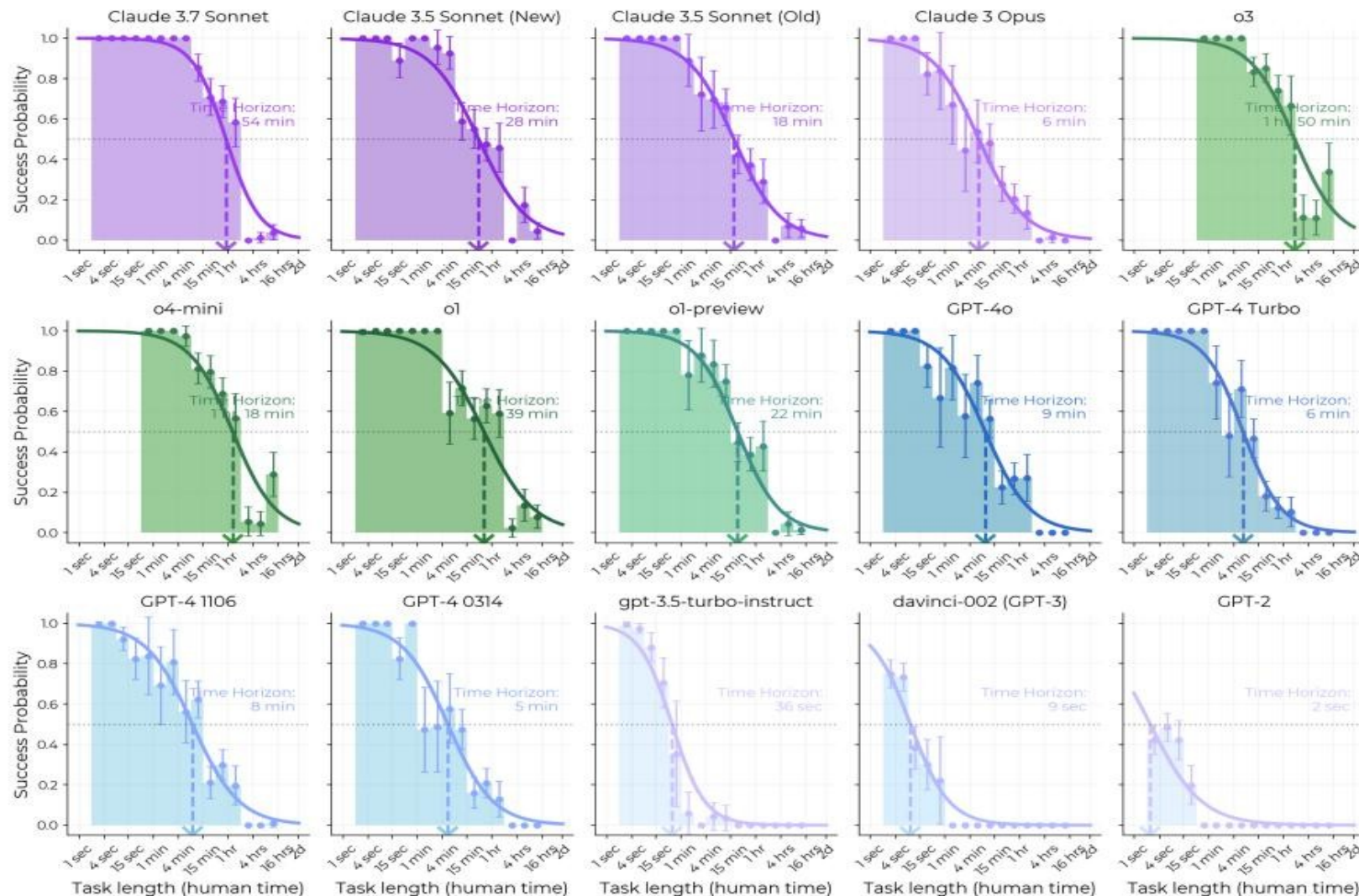
SWAA

66 tasks · 1 sec – 30 sec · Short single-step atomic actions

Task Name	Suite	Human Time	Description
find_shell_script	SWAA	3 seconds	Multiple choice: "Which file is a shell script?"
wikipedia_research	HCAST	1 minute	Research simple factual information from Wikipedia
oxdna_simple	HCAST	9 minutes	Detect and fix a bug in molecular dynamics simulation input files
munge_data	HCAST	56 minutes	Write a Python script to transform JSON data by inferring conversion rules
cuda_backtesting	RE-Bench	8 hours	Implement custom CUDA kernels for 30× speedup of a backtesting tool

AI Task Horizon: 2 Seconds (GPT-2, 2019) → ~1 Hour 50 Min (o3, 2025)

50% task-completion time horizon vs. model release date | Doubling time $\approx$ 207 days | $R^2 = 0.97$



207 days

Doubling time

$R^2 = 0.97$

Trend fit quality

~2 sec

GPT-2 (2019)

~5 min

GPT-4 (0314, 2023)

~1 hr 50 min

o3 (2025)

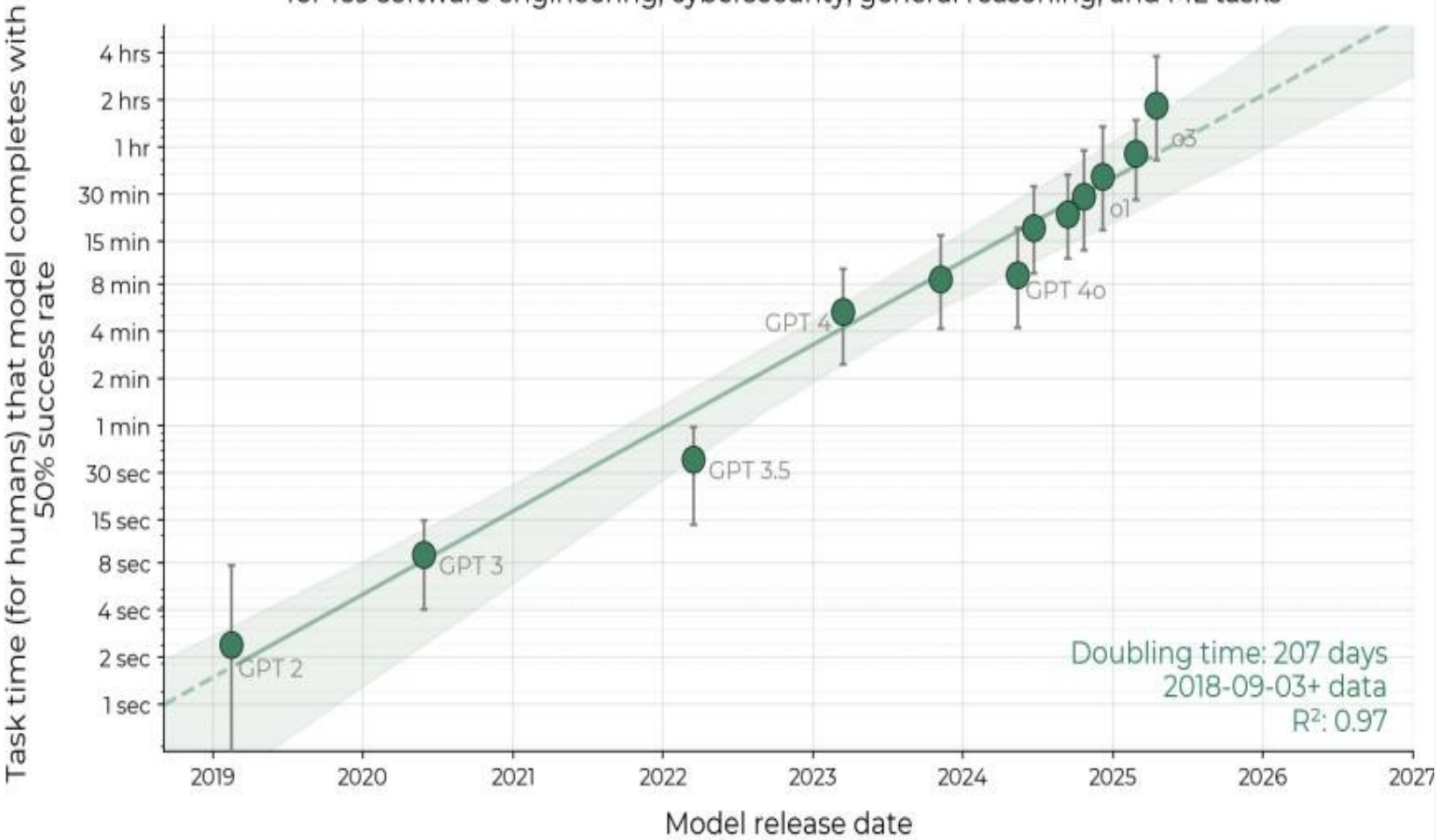
+20%

Faster growth 2023–2025
vs. 2019–2025 overall

Success Probability Drops Sharply as Task Length Increases for All Models

Frontier models (Claude 3.7 Sonnet, o3) push the curve rightward; older models plateau early

Length of tasks AI agents have been able to complete autonomously for 169 software engineering, cybersecurity, general reasoning, and ML tasks



Model 50% Horizons

~2 sec	GPT-2 (2019)
~9 sec	GPT-3 davinci (2020)
~36 sec	GPT-3.5-turbo (2023)
~5 min	GPT-4 (0314) (2023)
~6 min	Claude 3 Opus (2024)
~9 min	GPT-4o (2024)
~28 min	Claude 3.5 Sonnet (2024)
~39 min	o1 (2024)
~54 min	Claude 3.7 Sonnet (2025)
~1 hr 50 m	o3 (2025)

80% horizon is ~5× shorter than 50% horizon across all models

Reasoning, Tool Use, and Error Recovery Drive Time Horizon Growth

Four key capability drivers; performance degrades on less-structured tasks

Logical Reasoning

Improved planning and multi-step inference allow models to decompose complex tasks without human guidance.

Tool Use

Better ability to call shell commands, code interpreters, and APIs with correct syntax and sequencing.

Reliability & Self-Awareness

Models detect when they are off-track, verify outputs, and avoid cascading errors that derail long tasks.

Adaptability to Mistakes

Ability to recover from partial failures, retry with corrected approaches, and reach task completion despite setbacks.

Key Limitations

- Performance is notably lower on less-structured, 'messier' tasks that lack clear success criteria.
- External validity remains an open question — the task suite may not perfectly represent average real-world intellectual labor.
- Supplementary experiments found little evidence that trends are slower on more realistic tasks, but the question is not settled.

Extrapolation Predicts Month-Long Task Automation by 2028–2031

Continuing the 207-day doubling trend from o3's ~1h50m horizon



Naive extrapolation: if doubling every 207 days continues from o3's ~1h50m, AI reaches ~167 work-hour (1-month) tasks between mid-2028 and mid-2031 — implying potential automation of many month-long software projects.

Caveats & Uncertainty

- The 2023–2025 growth rate is ~20% faster than the overall 2019–2025 rate — possible acceleration.
- Trend could slow due to benchmark saturation, harder task distributions, or capability plateaus.
- External validity of the task suite to real-world job distributions remains an open empirical question.

Takeaways: Exponential Autonomous Capability Growth Demands Proactive Safety Governance

1

Robust Metric Established

The 50% time horizon provides an intuitive, human-grounded metric for tracking AI autonomy. $R^2 = 0.97$ over 6 years of model releases confirms the exponential trend is not noise.

2

Doubling Every ~7 Months — Possibly Accelerating

The ~207-day doubling time held from 2019–2025, with the 2023–2025 rate ~20% faster. Policy timelines must account for continued or accelerating growth.

3

Frontier Capability: ~2 Hours of Human-Equivalent Work (o3)

Current state-of-the-art agents autonomously complete software tasks equivalent to nearly two hours of skilled human labor — up from 2 seconds in 2019.

4

Month-Long Automation on the Horizon (2028–2031)

Extrapolation suggests AI may autonomously execute month-long software projects within 3–6 years, with major implications for workforce automation and AI safety oversight.

5

Limitations Require Ongoing Evaluation

External validity, poor performance on unstructured tasks, and uncertain generalization to real-world job distributions mean continued empirical measurement is essential — not optional.